Individual Borrower Risk Prediction and Kolmogorov–Smirnov (KS) Statistic

INTRODUCTION :

In machine learning–based credit risk assessment, evaluating the performance of a model is crucial to ensure reliable predictions. Financial institutions rely on such models to determine whether a borrower is likely to repay a loan or default. Proper evaluation helps determine how well the model distinguishes between high-risk and low-risk borrowers.

This project uses a Random Forest classification model to predict loan default risk. To analyze the model's performance and its practical usability, two key components were implemented:

1. Individual Borrower Risk Prediction
2. Kolmogorov–Smirnov (KS) Statistic for model evaluation

These methods help assess both model performance and real-world applicability.

INDIVIDUAL BORROWER RISK PREDICTION :

The purpose of this section is to demonstrate how the trained model can be used to predict the default risk for an individual loan applicant.

This simulation mimics real banking systems where financial institutions evaluate a borrower’s financial profile before approving a loan.

METHODOLOGY :

To simulate an individual borrower, in the first line of code snippet, a new sample applicant is created using the average values of all features in the dataset. This generates a realistic baseline borrower profile.

```python
# Predict risk for a single applicant using only credit score

# Create a row with average values for all features
new_person = X.mean().to_frame().T

# Change only the credit score
new_person["credit_score"] = 500   # you can change this value

# Predict risk
prediction = model.predict(new_person)
probability = model.predict_proba(new_person)[:,1]

print("Credit Score:", new_person["credit_score"].values[0])
print("Default Probability:", probability[0])

if prediction[0] == 1:
    print("Risk Level: HIGH RISK")
else:
    print("Risk Level: LOW RISK")
```

After creating this baseline profile, the credit score is modified to observe how the model responds to different credit risk levels, The user can change the credit score to check the risk level using the borrower's credit score, and further identify risk level's for the loan amount.

The trained Random Forest model then predicts:

- Whether the borrower is high risk or low risk
- The probability of default

OUTPUT INTERPRETATION :

The model produces two outputs:

Default Probability
This represents the likelihood that the borrower will default on the loan.

RISK PREDICTION	MEANING
HIGH RISK	Borrower likely to default
LOW RISK	Borrower likely to repay

EXAMPLE OUTPUT :

```
Credit Score: 500
Default Probability: 0.27
Risk Level: LOW RISK
```

This means that a borrower with a credit score of 500 has a low probability of default, and therefore, the model classifies the borrower as low risk.

KOLMOGOROV–SMIRNOV (KS) STATISTIC:

The KS Statistic is used to evaluate how effectively the model distinguishes between defaulters and non-defaulters, It measures the maximum difference between the cumulative probability distributions of these two groups. This metric is widely used in credit risk modeling and banking analytics.

METHODOLOGY :

The predicted probabilities are separated into two groups:

- Borrowers who defaulted
- Borrowers who did not default

```
from scipy.stats import ks_2samp

# Separate probabilities for each class
prob_default = y_prob[y_test == 1]
prob_non_default = y_prob[y_test == 0]
```

The Kolmogorov–Smirnov test is then used to compare these distributions

```
# Compute KS statistic
ks_statistic = ks_2samp(prob_default, prob_non_default)

print("KS Statistic:", ks_statistic.statistic)
```

OUTPUT INTERPRETATION :

```
KS Statistic: 0.34761009667024706
```

INTERPRETATION OF KS VALUES :

The ks value's for model are classified as :

KS VALUE'S	INTERPRATATION
Value's Less than 0.20	Poor model
Value's Ranging between 0.20-0.40	Moderate Model
Value's Ranging between 0.40-0.60	Good Model
Value's Above 0.60	Strong Model

A higher KS value indicates that the model is better at separating high-risk borrowers from low-risk borrowers.

For the given model, Since ks value was 0.38, we can conclude that the model shows moderate discrimination ability in separating defaulters from non-defaulters.

FINAL SUMMARY :

The implemented methods demonstrate how machine learning can be applied to evaluate credit risk effectively. The individual borrower risk prediction component shows how the trained model can estimate the probability of loan default for a specific applicant. Meanwhile, the KS statistic provides a quantitative measure of how well the model differentiates between borrowers who are likely to default and those who are likely to repay.